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End Semester Examination 2025

Name of the Program: B.Tech. Biotechnology
Name of the Course: Plant Biotechnology

Semester: VI
Course Code: TBT-602

Time: 3Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1
(a)	Explain how Plant Tissue Culture (PTC) works, and describe its advantages and limitations.	
(b)	Compare the direct and indirect methods of plant tissue culture with suitable example.	
(c)	What is somaclonal variation? Explain the molecular basis of somaclonal variation.	
Q2	(20marks)	CO2
(a)	Analyze the advantages and limitations of Androgenesis and Gynogenesis in the context of haploid plant production.	
(b)	How are protoplast isolation and fusion performed? Discuss the advantages and limitations of this technique.	
(c)	What is synthetic seed? Explain the method of artificial seed production.	
Q3	(20marks)	CO3
(a)	Assess the impact of Forest Biotechnology on the conservation of endangered plant species.	
(b)	Write short notes on 1. Cryopreservation 2. Green revolution	
(c)	What is Nitrogen fixation? Assess the role of nitrogen fixation in agricultural systems.	
Q4	(20marks)	CO4
(a)	What is Genetic transformation? Explain the methods of genetic transformation.	
(b)	Evaluate the effectiveness of <i>Agrobacterium rhizogenes</i> in the genetic transformation of plants.	
(c)	Write short notes on 1. Binary vector 2. <i>Agrobacterium tumefaciens</i>	
Q5	(20marks)	CO5
(a)	Explain the role of callus suspension culture in Secondary metabolite production with suitable example.	
(b)	Discuss social and legal issues associated with transgenic plants.	
(c)	Write short notes on 1. BT Crops 2. Secondary metabolites	

